Supplemantray Material

Climate Policy Uncertainty and Regime-Dependent Stock Return Predictability

This appendix presents additional material that underpins the main empirical analysis. It contains comprehensive model selection outcomes, forecasting performance over the full sample and during crisis periods, and statistical comparisons based on the Diebold–Mariano and Clark–McCracken tests. The purpose of the appendix is to strengthen transparency and reproducibility by providing full estimation results, robustness analyses, and alternative significance levels cited in the main text.

3. Methodology

Forecasting stock returns is difficult because predictive relationships are weak, unstable, and time-varying. As a result, the literature emphasizes comparative, out-of-sample forecasting rather than in-sample fit or a single preferred model. Early work shows that linear forecasting relations often break down over time, motivating systematic model comparisons within a common information set (Goyal and Welch, 2008; Rapach et al., 2010). More recent studies find that machine-learning methods can sometimes exploit nonlinear predictor interactions to improve forecasts, but their performance is highly regime-dependent and often no better than simple benchmarks (Gu et al., 2020). Consequently, researchers have proposed hybrid and forecast-combination strategies that blend linear and nonlinear components to enhance robustness, rather than seeking models that work well in all market environments (Bianchi et al., 2021).

This study compares three forecasting approaches—linear econometric models, nonlinear machine-learning methods, and a hybrid specification—from an agnostic perspective, without imposing a ranking. Using the same information set, it examines the predictability of daily S&P 500 returns. Predictors include systematic risk factors, market liquidity measures, technical indicators of investor behavior, and climate policy uncertainty, capturing both long- and short-run return dynamics. The empirical design focuses on out-of-sample performance, robustness during financial crises, and formal tests of predictive accuracy, recognizing the low signal-to-noise ratio in daily returns. Model comparisons use loss-based criteria and the Diebold–Mariano (1995) test for general forecast comparison, complemented by the Clark–McCracken (2001, 2005) tests for nested model evaluation, ensuring statistically valid inference when competing models share overlapping information sets.

We employ three forecasting approaches. First, an ARIMAX model captures linear relationships, factor loadings, and liquidity effects in a transparent econometric framework. Second, a Long Short-Term Memory (LSTM) neural network models nonlinear, path-dependent dynamics linked to behavioral trading, volatility clustering, and regime changes. Third, a hybrid ensemble combines ARIMAX and LSTM forecasts, retaining linear structure while adding nonlinear adaptability to structural breaks and uncertainty shocks. We evaluate performance using a time-ordered training–validation–test split and reassess it across multiple crisis episodes, enabling consistent comparison under both normal markets and periods of elevated financial and policy uncertainty.

3.1 Data and Variable Construction

The empirical analysis uses daily S&P 500 data and a broad set of explanatory variables capturing systematic risk, market liquidity, technical trading, and climate-policy uncertainty. The long sample includes major crises—the Global Financial Crisis, the Eurozone debt crisis, the oil price collapse, the COVID-19 shock, the 2022 inflation-driven monetary tightening, and the Russia-Ukraine and Israel-Palestine conflicts—to examine whether stock return predictability and the importance of predictive features differ across market regimes. Daily returns are computed as logarithmic differences of closing prices:

$$r_t = \ln(P_t) - \ln(P_{t-1}), \quad (1)$$

where P_t denotes the closing level of the S&P 500 index on trading day t. This transformation is standard in the high-frequency financial econometrics literature and ensures that the resulting series possesses time-additive properties.

The Fama–French four-factor (FF4) model captures systematic risk through size (SMB), value (HML), profitability (RMW), and investment (CMA) (Fama and French, 2015). The market factor (MKT–RF) is excluded to avoid mechanical correlation with the benchmark index, which would inflate in-sample fit and undermine out-of-sample performance evaluation. The momentum factor (MOM) is added separately to assess whether short-term return patterns persist or reverse. The Amihud illiquidity measure is an empirical proxy for market liquidity, capturing the sensitivity of asset prices to trading volume and changes in market depth (Amihud, 2002). Common technical indicators for tracking market behavior and volatility include the 10-day moving average (MA_{10}), the 14-day Relative Strength Index (RSI_{14}), the Moving Average Convergence Divergence (MACD), and Bollinger Band Width (BB_Width). At a daily frequency, these indicators describe trend strength, extremes in market sentiment, shifts in price momentum, and changes in volatility regimes.

The Climate Policy Uncertainty (CPU) index quantifies uncertainty about climate-related regulation and transition policies, capturing changes in market expectations about the timing, severity, and design of policy interventions. Consistent with recent empirical studies, CPU is incorporated as an informational predictor rather than a priced risk factor, reflecting its documented influence on stock market dynamics and forecasting performance under conditions of regulatory and transition-related uncertainty (Chen et al., 2023; Raza et al., 2024). Prior research links climate policy uncertainty to equity market behavior via uncertainty and expectations channels, affecting stock returns, volatility, and investor decisions (Gavriilidis, 2021; Treepongkaruna et al., 2023). Recent work shows that climate policy uncertainty also helps predict short-term financial market movements, particularly during periods of market stress and regulatory ambiguity (Liang et al., 2022; Raza et al., 2024). For empirical analysis, the CPU index is interpolated to daily frequency to match the return data and used to test whether climate policy uncertainty improves short-horizon stock return forecasts in both normal and stressed markets. All variables are aligned by trading day, and only information available at time t is used to predict returns at t+1 to avoid look-ahead bias.

3.2 Research Design and Implementation

To rigorously assess return predictability, the empirical design uses a strictly chronological forecasting framework. The sample is split into three non-overlapping subsamples: a training set for model estimation, a validation set for hyperparameter tuning and early stopping, and a test set for out-of-sample performance evaluation. This temporal partitioning prevents information leakage and better reflects real-time conditions faced by investors and risk

managers. All explanatory variables are standardized via min–max scaling, fitted only on the training sample and then applied to validation and test samples to avoid look-ahead bias. For the LSTM, input data are organized into rolling windows of fixed length so the network can learn temporal dependencies in past returns and predictors. A 30-trading-day window is used to balance capturing short-term dynamics against over-parameterization.

Missing observations occur because technical indicators need initial lookback windows. These early observations are excluded from forecasting to avoid artificial or look-ahead bias. All models use only information available up to time *t* and then generate one-step-ahead forecasts for period *t+1*. Crisis-specific evaluations apply the optimal model specifications from the full-sample analysis to subsamples for each crisis. Parameters are estimated using only pre-crisis data and then evaluated over the crisis window. This tests the robustness of predictions under abrupt regime shifts and isolates the effects of structural breaks and uncertainty shocks on forecasting performance from the influence of crisis-period data on parameter estimation. This research design jointly estimates, validates, and evaluates models in empirically grounded forecasting scenarios, enabling systematic, methodologically consistent comparisons of linear, nonlinear, and hybrid approaches in both stable and high-volatility markets.

3.3 Model Specification

This study uses three complementary forecasting frameworks—ARIMAX, LSTM, and a hybrid ensemble—each capturing different aspects of return dynamics. ARIMAX provides a transparent linear model with lagged dependencies and key explanatory variables. LSTM is designed to uncover nonlinear, path-dependent structures, especially under rapid regime shifts and high uncertainty. The hybrid model combines ARIMAX's systematic, factor-based forecasts with LSTM's flexible modeling of complex, regime-dependent behavior.

ARIMAX Model: The Autoregressive Integrated Moving Average with Exogenous Variables (ARIMAX) model extends the classical ARIMA framework by capturing both serial autocorrelation in the dependent variable (e.g., asset returns) and the effects of external covariates or regressors (Box & Jenkins, 1970). Its general form is:

$$(1 - L)r_t = \alpha_0 + \sum_{i=1}^{p} \phi_i (1 - L)r_{t-i} + \beta' x_t + \epsilon_t \quad (2)$$

where:

- r_t is the daily return at time t,
- L is the lag operator,
- ϕ_i are the autoregressive coefficients,
- x_t represents the external regressors (FF4 factors, liquidity, CPU, technical indicators),
- ϵ_t is the white noise error term.

LSTM Model: The Long Short-Term Memory (LSTM) architecture (Hochreiter and Schmidhuber, 1997) is used to model long-range temporal dependencies and complex non-linear relationships in financial time series. LSTM networks learn temporal patterns through gating mechanisms that control information flow, allowing the model to selectively retain or discard information from previous time steps. Formally, the LSTM model is defined by the following equations:

Forget gate:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (3)$$

Input gate:

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (4)$$
$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c) \quad (5)$$

Cell state:

$$c_t = f_t \cdot c_{t-1} + i_t \cdot \tilde{c}_t \quad (6)$$

Output gate:

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \quad (7)$$
$$h_t = o_t \cdot \tanh(c_t) \quad (8)$$

The final prediction is given by:

$$\hat{r}_{t+1} = W_f h_t + b \quad (9)$$

where h_t is the output of the LSTM at time t, σ is the sigmoid activation function, and W_f, W_i, W_o, W_c are the weight matrices for the forget, input, output, and cell state, respectively.

Hybrid Model: The hybrid model integrates the most effective components of ARIMAX and LSTM (Zhang, 2003; Zhang et al., 2020), aiming to offset each model's limitations by leveraging their complementary strengths. The ensemble prediction is obtained by linearly combining the ARIMAX and LSTM outputs, assigning each forecast a distinct weight:

$$\hat{r}_t^{Hybrid} = \gamma_1 \hat{r}_t^{ARIMAX} + \gamma_2 \hat{r}_t^{LSTM} \quad (10)$$

where γ_1 and γ_2 are the model weights that were found by training on the validation set. This method lets the model mix linear, factor-driven predictions with non-linear, time-varying behaviors in a way that works best for it.

Model Selection and Evaluation: We use a training–validation–test split to assess out-of-sample predictive performance. The training set estimates model parameters, the validation set tunes hyperparameters and implements early stopping (especially for LSTM models), and the test set is used for final performance evaluation. We also apply the Diebold–Mariano (DM) test to compare models' predictive accuracy using loss differentials based on squared prediction errors.

3.4 Evaluation Metrics

We assess forecast quality using standard error-based and explanatory performance metrics common in return predictability studies. Because daily equity returns have a low signal-to-noise ratio, we focus on out-of-sample predictive performance rather than in-sample fit. To assess forecast accuracy, we use Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE) as primary metrics.

$$RMSE = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(r_t - \hat{r}_t)^2}, \quad MAE = \frac{1}{T}\sum_{t=1}^{T} | r_t - \hat{r}_t |, \tag{11}$$

where r_t denotes the realised return and $\hat{r}_t$ is the expected return. RMSE punishes big mistakes in forecasts more than MAE does, which is a good way to measure average absolute deviation. These metrics are especially useful during times of crisis, when return volatility and forecast dispersion tend to go up. Second, the out-of-sample coefficient of determination (R^2) is used to compare the explanatory power of different models to a historical mean benchmark:

$$R^2 = 1 - \frac{\sum_{t=1}^{T}(r_t - \hat{r}_t)^2}{\sum_{t=1}^{T}(r_t - \bar{r})^2}, \tag{12}$$

where $\bar{r}$ is the average return for the time period being looked at. Negative out-of-sample R^2 values mean that the benchmark does better than the forecasting model, which is a common result in daily return forecasting. The Diebold–Mariano (DM) test formally compares the predictive accuracy of competing forecasting models (Diebold and Mariano, 1995). It evaluates whether the expected loss differential between two forecasts differs significantly from zero. Here, the loss function is the squared forecast error, and the null of equal predictive accuracy is tested against the alternative that one model has systematically lower forecast errors. This provides an inferential complement to descriptive performance measures and is used to establish a statistically rigorous ranking of ARIMAX, LSTM, and hybrid forecasting models.

Statistical differences in predictive accuracy are evaluated using the Diebold and Mariano (1995) test based on squared-error loss differentials. Let $L(\cdot)$ denote the loss function and define the loss differential between models i and j as

$$d_t = L(e_{i,t}) - L(e_{j,t}), \qquad e_{k,t} = r_t - \hat{r}_{k,t}. \tag{13}$$

Under the null of equal predictive accuracy, $E[d_t] = 0$. The Diebold–Mariano statistic is

$$DM = \frac{\bar{d}}{\sqrt{\widehat{Var}(\bar{d})}}, \ \bar{d} = \frac{1}{T}\sum_{t=1}^{T} d_t, \tag{14}$$

where $\widehat{Var}(\bar{d})$ is a heteroskedasticity- and autocorrelation-consistent estimator. We implement the test using squared-error loss.

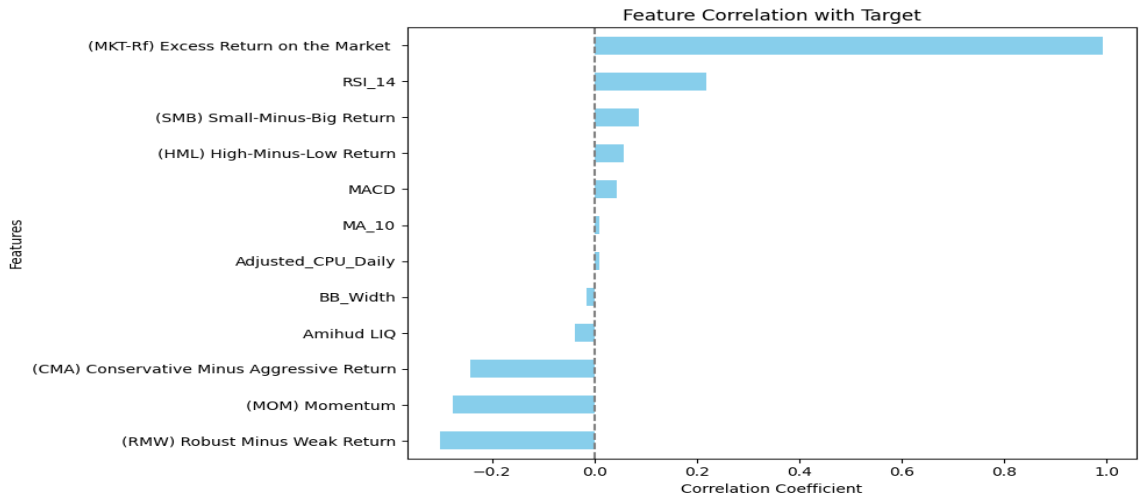
Diebold–Mariano tests are reported for all pairwise comparisons. For nested or partially encompassing models, inference follows Clark–McCracken (2001, 2005) framework, which is valid under nesting. This is crucial because the Hybrid specification combines ARIMAX and LSTM forecasts, so the competing forecasts are dependent and the Diebold–Mariano test may exhibit size distortions. We therefore use two Clark–McCracken statistics: the ENC-NEW test, which assesses whether the larger model adds predictive information beyond the benchmark, and the MSE-F test, which tests whether the alternative model significantly reduces mean squared forecast error. Both are designed for nested models and are more appropriate when models share overlapping information sets.

4. Results and Discussion

4.1 Feature Behaviour in the Full Sample

This subsection examines the explanatory variables in the full sample to benchmark which features are informative under normal market conditions. Before evaluating model performance, it is necessary to analyse correlations among predictors and between predictors and daily S&P 500 returns to identify relevant information and avoid mechanical dependencies that could bias forecasts. Figure 1's correlation heatmap shows that the market excess return (MKT–RF) is almost perfectly correlated with the index return because the dependent variable is constructed from market prices. This mechanical overlap inflates the model's apparent explanatory power and obscures the role of other covariates. Following standard practice in index-level return forecasting, MKT–RF is therefore excluded from further analyses to preserve interpretability and allow a meaningful assessment of out-of-sample performance.

Figure S1: Full-sample correlation coefficients (with MKT–RF)



After removing the market factor (MKT–RF), Figure 2 reveals the remaining correlation structure. With this source of contamination eliminated, economically meaningful relationships become clearer: profitability (RMW) and investment (CMA) are the Fama–French factors most strongly associated with daily returns, while size (SMB) and value (HML) are more weakly related. Table 1 quantifies these patterns, showing that signals from fundamental factor models still explain daily returns once mechanical market dependence is controlled for. Momentum (MOM) shows a strong negative association with returns, consistent with short-term reversal effects in equity index data.

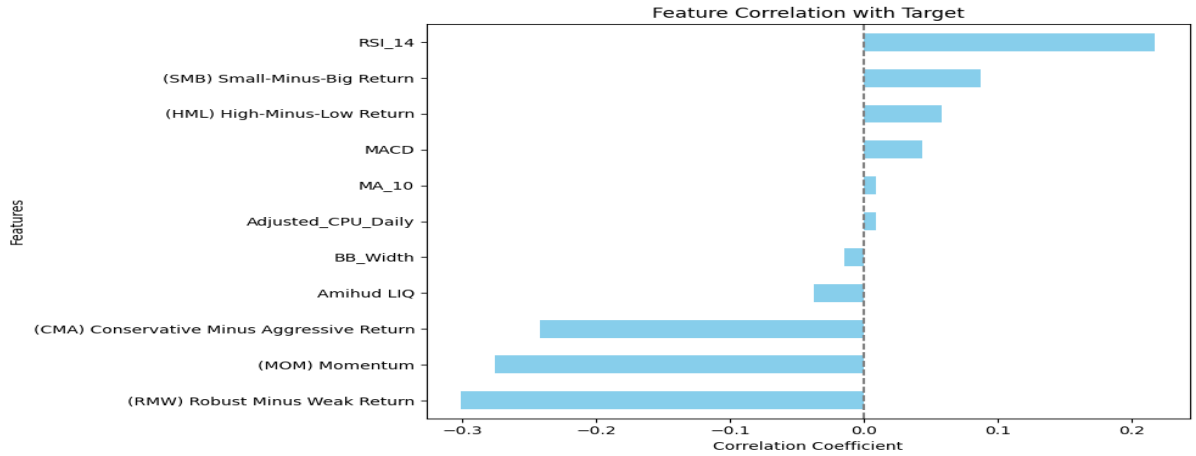
Table S1: Strongest Positive and Negative Correlations with Daily Returns (Whole Sample, Excluding MKT–RF)

Rank	Variable	Correlation	Rank	Variable	Correlation
Positive			**Negative**		
1	RSI_{14}	**0.2176**	**1**	RMW	**–0.3005**
2	SMB	0.087	**2**	MOM	–0.2754
3	HML	0.0577	**3**	CMA	–0.2421
4	MACD	0.0435	**4**	Amihud Illiquidity	–0.0372

5	MA_{10}	0.0094	**5**	Bollinger Band Width	–0.0150

Liquidity conditions are also important in the full-sample analysis. Figure 2 and Table 1 show that the Amihud illiquidity measure is consistently negatively related to daily returns. As market depth deteriorates, contemporaneous returns fall even in non-crisis periods. This supports viewing liquidity as a persistent state variable that systematically affects return dynamics, rather than as an indicator active only during market distress. Technical indicators capture aggregate investor behavior. Among them, the Relative Strength Index (RSI_{14}) is most strongly positively correlated with other measures, followed by the short-term moving average (MA_{10}) and MACD. As shown in Table 1, these relationships suggest that sentiment imbalances, short-horizon trend persistence, and moderate momentum help explain return variation in normal markets. In contrast, Bollinger Band Width is only weakly correlated over the full sample, indicating that volatility-regime indicators add little explanatory power except during periods of severe market stress.

Figure 2: Full-sample correlation coefficients excluding MKT–RF



Climate Policy Uncertainty (CPU) also shows a modest but systematic link with daily stock returns. As Figure 2 and Table 1 report, this link is weaker than that of factor and liquidity variables but remains statistically significant. This suggests that uncertainty about climate- and environment-related policy is priced even in non-crisis periods and provides preliminary evidence that climate policy risk affects short-term market dynamics absent extreme events. The full-sample feature analysis identifies a core set of informative predictors under standard market conditions. Figures 1–2 and Table 1 show that the FF4 factors—especially RMW and CMA—along with momentum, liquidity, technical indicators (RSI_{14}, MACD, MA_{10}), and climate-policy uncertainty, have strong explanatory power. These variables constitute the main information set for assessing model performance and selecting optimal forecasting specifications in the next section.

4.2 Whole-Sample Model Performance and Identification of Best-Performing Specifications

This subsection evaluates the models' predictive performance over the full sample and selects the best specifications using forecast accuracy—measured by RMSE and MAE—and out-of-sample explanatory power. The analysis considers three feature sets: (i) a baseline with the Fama–French and momentum factors, (ii) an extension adding liquidity and climate policy

uncertainty, and (iii) a full specification that also includes technical indicators. For each specification, performance is evaluated on the training, validation, and test sets to ensure results reflect true predictive ability rather than in-sample overfitting.

4.2.1 Baseline Specification: FF4 and Momentum

Table 2 reports baseline results using the FF4 factors (SMB, HML, RMW, CMA) plus momentum (MOM). The LSTM yields the lowest in-sample RMSE (0.01257) and MAE (0.00853) and a positive out-of-sample R^2 of 0.0203, indicating that modeling non-linear temporal dependencies adds predictive power even with only factor-based variables. In contrast, the ARIMAX model has larger forecast errors and a non-positive out-of-sample R^2, suggesting linear dynamics alone cannot capture short-horizon return fluctuations.

Table S2: Baseline model performance (FF4 & MOM)

Baseline Training Performance (Training)				
Model	Features	RMSE	MAE	R^2
ARIMAX	SMB + HML + RMW + CMA + MOM	0.012204	0.008341	0.0798
LSTM	SMB + HML + RMW + CMA + MOM	0.012572	0.008530	0.0203
Hybrid	SMB + HML + RMW + CMA + MOM	0.012081	0.008060	0.1085
Baseline Validation Performance (Valid)				
Model	Features	RMSE	MAE	R^2
ARIMA**X**	SMB + HML + RMW + CMA + MOM	0.013245	0.009239	–0.1398
LSTM	SMB + HML + RMW + CMA + MOM	0.013203	0.009183	–0.1362
Hybrid	SMB + HML + RMW + CMA + MOM	0.013017	0.008954	–0.1031
Baseline Test Performance (Test)				
Model	Features	RMSE	MAE	R^2
ARIMA**X**	SMB + HML + RMW + CMA + MOM	0.01475	0.00994	–0.2076
LSTM	SMB + HML + RMW + CMA + MOM	0.01365	0.00898	–0.0207
Hybrid	SMB + HML + RMW + CMA + MOM	0.01383	0.00911	–0.0481

The hybrid model delivers slightly better forecast performance than the LSTM in the validation sample, based on forecast error metrics, even though both models have negative R^2 values. In the test sample, however, the LSTM again achieves the lowest RMSE (0.01365) and MAE (0.00898), indicating that its nonlinear architecture outperforms both the linear and ensemble alternatives in low-information settings. Overall, the LSTM with FF4 and momentum factors is the preferred baseline specification.

4.2.2 Extended Specification: Liquidity and Climate Policy Uncertainty

Table 3 extends the feature set with the Amihud illiquidity measure and CPU index, producing clear gains in predictive accuracy, especially for the hybrid model. In-sample, the hybrid specification achieves lower RMSE (0.01155) and MAE (0.00794), and a higher R^2 of 0.1868. These results show that liquidity frictions and policy uncertainty add useful information that factor-based hybrid signals can exploit, while purely linear or purely non-linear models cannot fully do so. The inclusion of climate policy uncertainty contributes incremental forecasting information in several model–regime combinations, particularly during periods characterized by heightened policy and market uncertainty. This finding is consistent with prior studies documenting that climate policy uncertainty affects stock market dynamics and volatility through regulatory ambiguity and transition risk channels (Chen et al., 2023; Treepongkaruna

et al., 2023). Importantly, the predictive contribution of climate policy uncertainty is not uniform across models, echoing recent evidence that its forecasting relevance depends on model design and market context (Raza et al., 2024; Liang et al., 2022). These results suggest that climate policy uncertainty operates as a state-dependent information variable rather than a constant source of predictive power, reinforcing the need to evaluate its role within multiple forecasting paradigms.

Table S3: Model performance with liquidity and CPU

Performance with liquidity and CPU (Train)				
Model	Features	RMSE	MAE	R^2
ARIMAX	SMB + HML + RMW + CMA + MOM + LIQ + CPU	0.011859	0.008263	0.1427
LSTM	SMB + HML + RMW + CMA + MOM + LIQ	0.012814	0.008625	–0.0009
LSTM	SMB + HML + RMW + CMA + MOM + LIQ + CPU	0.012866	0.008617	–0.0091
Hybrid	SMB + HML + RMW + CMA + MOM + LIQ + CPU	0.011550	0.007938	0.1868
Performance with liquidity and CPU (Valid)				
Model	Features	RMSE	MAE	R^2
ARIMAX	SMB + HML + RMW + CMA + MOM + LIQ	0.009056	0.006547	–0.1499
ARIMA**X**	SMB + HML + RMW + CMA + MOM + LIQ + CPU	0.009060	0.006552	–0.1508
LSTM	SMB + HML + RMW + CMA + MOM + LIQ + CPU	0.008478	0.005834	–0.0078
Hybrid	SMB + HML + RMW + CMA + MOM + LIQ + CPU	0.008571	0.006076	–0.0299
Performance with liquidity and CPU (Test)				
Model	Features	RMSE	MAE	R^2
ARIMA**X**	SMB + HML + RMW + CMA + MOM + LIQ	0.014749	0.009934	–0.1922
ARIMA**X**	SMB + HML + RMW + CMA + MOM + LIQ + CPU	0.014754	0.009934	–0.1930
LSTM	SMB + HML + RMW + CMA + MOM + LIQ + CPU	0.013646	0.008983	–0.0206
Hybrid	SMB + HML + RMW + CMA + MOM + LIQ + CPU	0.013786	0.009080	–0.0415

The hybrid model achieves the lowest validation errors (RMSE = 0.00857; MAE = 0.00608), outperforming both ARIMAX and LSTM. It is also more stable across subsamples, though R^2 remains negative for all models. Performance deteriorates in the test sample due to noisier out-of-sample returns, but the hybrid model stays competitive and avoids the sharp error increase seen in ARIMAX. These findings indicate that liquidity and climate policy uncertainty are regime-dependent predictors whose information is best captured by combining multiple models rather than relying on a single linear or non-linear specification.

4.2.3 Full Specification: Factors, Liquidity, CPU and Technical Indicators

Table 4 presents results for the full feature set, including technical indicators of short-term price dynamics and volatility. In the training sample, the hybrid model with MA_{10}, MACD, RSI_{14}, and Bollinger Band Width delivers the lowest forecasting errors (RMSE = 0.01107; MAE = 0.00768) and the highest explanatory power ($R^2 = 0.2533$). This in-sample performance shows that the hybrid framework effectively combines linear factor information with nonlinear trend-following and volatility-sensitive signals.

Table S4: Model performance with full feature set (FF4-factors, liquidity, CPU, technical indicators)

Performance with full feature set (Train)				
Model	Features	RMSE	MAE	R^2

ARIMAX	SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10} + BBWidth + MACD + RSI_{14}	0.011167	0.007878	0.2398
LSTM	SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10} + BBWidth	0.012811	0.008651	–0.0005
LSTM	SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10} + BBWidth + MACD + RSI_{14}	0.012829	0.008649	–0.0033
Hybrid	SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10} + BBWidth + MACD + RSI_{14}	0.011067	0.007684	0.2533
Performance with full feature set (Valid)				
Model	Features	RMSE	MAE	R^2
ARIMA**X**	SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10} + BBWidth + MACD + RSI_{14}	0.008866	0.006389	–0.1020
LSTM	SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10}	0.008506	0.005861	–0.0144
LSTM	SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10} + BBWidth + MACD + RSI_{14}	0.008562	0.005943	–0.0279
Hybrid	SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10} + BBWidth + MACD + RSI_{14}	0.008514	0.006056	–0.0162
Performance with full feature set (Test)				
Model	Features	RMSE	MAE	R^2
ARIMA**X**	SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10}	0.015178	0.010335	–0.2625
ARIMA**X**	SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10} + BBWidth + MACD	0.015067	0.010271	–0.2442
ARIMAX	**SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10} + BBWidth + MACD + RSI_{14}**	**0.014754**	**0.009934**	**–0.1930**
LSTM	**SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10} + BBWidth**	**0.013624**	**0.008961**	**–0.0173**
LSTM	SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10} + BBWidth + MACD + RSI_{14}	0.013646	0.008983	–0.0206
Hybrid	SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10} + BBWidth	0.014162	0.009536	–0.0991
Hybrid	**SMB + HML + RMW + CMA + MOM + LIQ + CPU + MA_{10} + BBWidth + MACD + RSI_{14}**	**0.013786**	**0.009080**	**–0.0415**

In the validation sample, the LSTM model with the parsimonious MA_{10}-only specification achieves the lowest forecasting errors (RMSE 0.00851, MAE 0.00586), slightly outperforming the hybrid configuration. This indicates that, over short validation horizons, the LSTM benefits from a reduced set of technical inputs, which limits overfitting while still capturing short-term trend persistence. In the test sample, the LSTM model again achieves the lowest forecast errors (RMSE = 0.01362; MAE = 0.00896) when combined with MA_{10} and Bollinger Band Width, while the hybrid model shows slightly higher errors and a more negative R^2. This highlights the trade-off between model complexity and generalisation: highly parameterised models can excel in-sample but may be less robust and stable out-of-sample.

4.2.4 Identification and Interpretation of Best-Performing Specifications

Overall, the results show that forecast accuracy depends on model class and feature set, with no single dominant specification. ARIMAX performs well in parsimonious setups but

deteriorates as markets become more nonlinear and regime-based. In contrast, LSTM and hybrid models gain more from adding technical indicators, liquidity measures, and climate policy uncertainty, indicating that nonlinear and composite architectures better capture short-horizon returns under changing market conditions. With a parsimonious yet informative set of technical indicators, the LSTM achieves comparable accuracy even with a relatively small or moderately engineered feature set. Its best configuration jointly uses FF4, MOM, liquidity, CPU, MA_{10}, and Bollinger Band Width, suggesting it effectively models market reactions to volatility shifts and short-horizon price trends. The hybrid model performs best when enhanced with additional liquidity measures, climate policy uncertainty (CPU), and more technical indicators. The optimal setup includes the Fama–French four-factor (FF4) model, a momentum (MOM) factor, liquidity, CPU, a 10-day moving average (MA_{10}), Moving Average Convergence Divergence (MACD), the 14-day Relative Strength Index (RSI_{14}), and Bollinger Band Width. This allows the model to exploit systematic factors, behavioral trading signals, and volatility dynamics simultaneously. Stability improves by combining linear and nonlinear forecasting components, lowering the risk of model misspecification. Based on these results, the following best-performing specifications are retained for subsequent analysis:

Model	Selected Specification	RMSE	MAE	OOS R^2
ARIMAX	FF4 + MOM + LIQ + CPU + MA_{10} + BBWidth + MACD + RSI_{14}	0.014754	0.009934	–0.1930
LSTM	FF4 + MOM + LIQ + CPU + MA_{10} + BBWidth	0.013624	0.008961	–0.0173
Hybrid	FF4 + MOM + LIQ + CPU + MA_{10} + BBWidth + MACD + RSI_{14}	0.013786	0.009080	–0.0415

These specifications are carried forward to the crisis-specific evaluation and formal statistical comparison using Diebold–Mariano tests. The optimal ARIMAX, LSTM, and hybrid model specifications are selected from these results and retained for further analysis. Overall, the hybrid framework delivers the strongest forecasting performance, particularly when momentum, liquidity measures, climate policy uncertainty, and technical indicators are jointly incorporated. Figures 3–5 show the correspondence between predicted and realised returns for the best-performing hybrid model in the training, validation, and test samples, respectively.

Figure S3: Model Fit in the Training Dataset: Actual and Predicted Returns

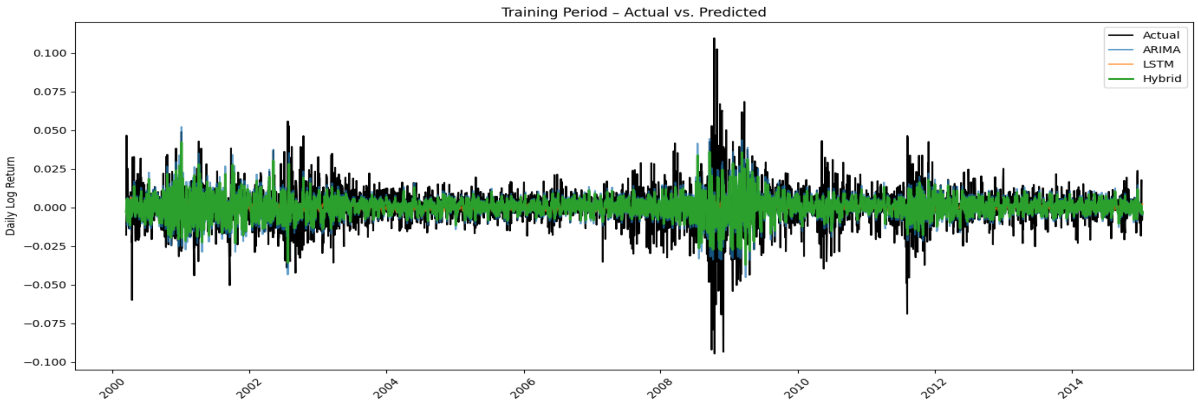


Figure S4: Validation Forecast Performance: Actual Versus Predicted Returns

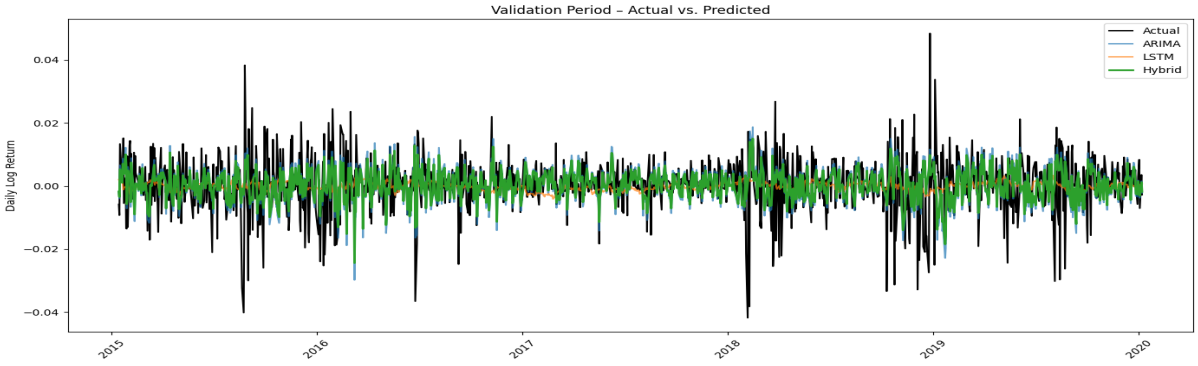
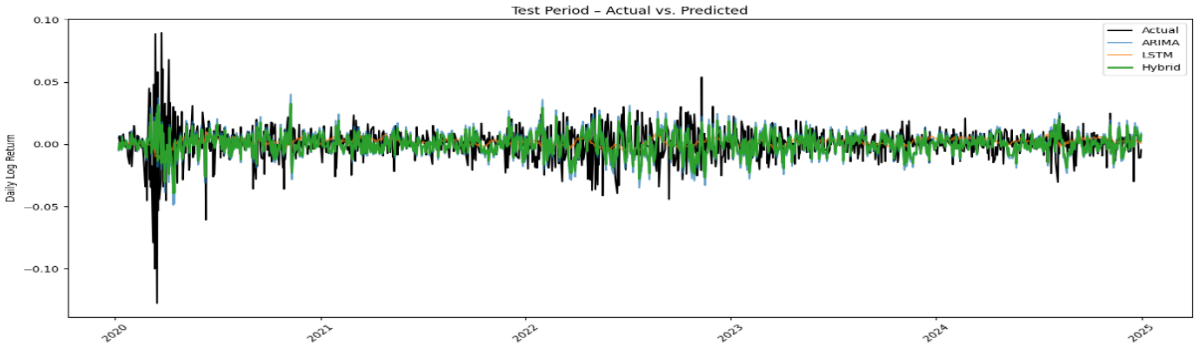


Figure S5: Test Sample Predictive Accuracy: Actual Versus Predicted Returns



4.2.5 Interpretation of Full-Sample Forecasting Results

The comparative forecasting evidence underscores a central result from the return predictability literature: performance depends heavily on the model used and on market conditions. In line with prior work on the instability of linear relations (Goyal and Welch, 2008; Rapach et al., 2010), the ARIMAX benchmark performs well in some subsamples but does not dominate out of sample. Nonlinear machine-learning models reveal additional predictive patterns in certain environments, consistent with studies showing that they exploit complex interactions among predictors (Gu et al., 2020; Coulombe et al., 2022), but they do not consistently outperform simpler benchmarks, especially in tranquil periods. Hybrid models, combining linear and nonlinear elements, deliver superior performance in some regimes but not uniformly, echoing evidence that they mainly enhance robustness rather than guarantee higher accuracy (Bianchi et al., 2021). Overall, no single forecasting framework dominates over time, highlighting the value of comparative, regime-sensitive forecast evaluation.

4.3 Crisis-Period Forecasting Performance

To assess how resilient forecasts are under extreme market conditions, we evaluate forecasting performance during crises using a pre-crisis estimation and crisis-period evaluation framework. Each model is estimated only with information available before a crisis, then used to generate forecasts throughout the crisis. This offers a stringent test of predictive ability, as model parameters are fixed before structural breaks and heightened uncertainty occur. We use the same loss functions and Diebold–Mariano tests as in the full-sample analysis to assess forecast accuracy during crises.

This section evaluates the stability and robustness of the top-performing ARIMAX, LSTM, and hybrid forecasting models during periods of severe financial market turmoil. Crises are marked by sharp volatility spikes, structural breaks, tighter liquidity, and heightened

macroeconomic and geopolitical uncertainty, all of which make return prediction more difficult. Assessing model performance in these stressed conditions is essential to determine whether the framework still delivers informative and reliable forecasts beyond relatively calm market regimes. The analysis covers seven major crises: the Global Financial Crisis, the Eurozone sovereign debt crisis, the oil price collapse, the COVID-19 pandemic, the inflationary shock period, the Russia–Ukraine conflict, and the Israel–Palestine conflict. Model performance is measured by RMSE, MAE, and out-of-sample R^2, enabling systematic comparison across models and crisis regimes.

Table S5**:** Comprehensive Out-of-Sample Forecast Performance Across the Full Test Period and Crisis Regimes

Period	ARIMA_RMSE	LSTM_RMSE	Hybrid_RMSE	ARIMA_MAE	LSTM_MAE	Hybrid_MAE	ARIMA_R2	LSTM_R2	Hybrid_R2	ARIMA_vs_Hybrid_Improvement_%	**LSTM_vs_Hybrid_Improvement_%**	**Crisis**
Full	0.055836854	0.013502001	0.013371541	0.045529821	0.009081581	0.00885964	-16.08639247	0.000909707	0.020123329	76.05248135	0.966223761	
GFC	0.030413813	0.026278879	0.023832049	0.023512465	0.019651912	0.016396425	-0.592992 32	-0.189275 98	0.021874 28	21.64070888	9.3107071	GFC
EuroDebt	0.014214 36	0.014593 08	0.011919227	0.011022935	0.011190553	0.008332757	-0.289056977	-0.358662 06	0.093612903	16.14657615	18.32274509	EuroDebt
OilCollapse	0.016346246	0.010656669	0.010188898	0.013410902	0.007923313	0.00748583	-1.398427803	-0.019373946	0.068152165	37.6682708	4.389467816	OilCollapse
COVID	0.054714063	0.022576265	0.022339022	0.041151401	0.014276479	0.014195763	-4.827386219	0.007843762	0.028558641	59.1713344	1.050853223	COVID
InflationTightening	0.075081394	0.015409653	0.014856463	0.069170868	0.012199494	0.011888862	-23.37161979	-0.026600998	0.045775383	80.21285615	3.589892446	InflationTightening
RussiaUkraine	0.028007455	0.015746264	0.015233425	0.023308045	0.012386377	0.012194197	-2.216569432	-0.016718423	0.048429975	45.60939226	3.25689322	RussiaUkraine
IsraelPalestine	0.037533433	0.007539863	0.007162208	0.033171478	0.005847297	0.005518189	-25.99837689	-0.089501785	0.016906132	80.91779023	5.008767994	IsraelPalestine

Notes: *RMSE and MAE measure forecast error, while out-of-sample R^2 evaluates explanatory power relative to the historical mean benchmark. Improvement percentages indicate the relative gain of the Hybrid model compared to ARIMAX and LSTM. Positive improvement values indicate better performance of the Hybrid model. Crisis periods correspond to major financial, macroeconomic, and geopolitical shocks as defined in the main text.*

Table S5 summarizes the out-of-sample forecasting performance of the selected ARIMAX, LSTM, and Hybrid specifications across the full test period and major crisis regimes. The results reveal clear regime dependence in predictive accuracy, with model performance varying systematically according to the nature, duration, and transmission mechanism of each crisis. Across all periods, the ARIMAX model performs worst, the LSTM model provides substantial improvements, and the Hybrid model delivers the most robust and generally strongest performance—particularly during crises characterized by persistent structural disruptions, policy uncertainty, and nonlinear market adjustments.

The ARIMAX model deteriorates sharply across all crisis periods. Forecast errors remain consistently high, while out-of-sample R^2 values are uniformly negative and often strongly so, indicating that the linear specification fails to outperform a simple benchmark under stress. This failure is most pronounced during the inflation tightening period and the Israel–Palestine conflict**,** where macro-financial conditions were dominated by abrupt repricing, elevated uncertainty, and unstable investor expectations. In such environments, linear autoregressive models—even when augmented with factor-based and uncertainty covariates—cannot adequately capture the nonlinear adjustments, feedback loops, and structural breaks that characterize crisis-driven markets. This pattern is consistent with earlier evidence showing that linear forecasting models often break down during episodes of financial instability and regime shifts (Guidolin & Timmermann, 2006).

The LSTM model consistently improves upon ARIMAX across all regimes, delivering substantially lower RMSE and MAE. This reflects its ability to capture nonlinear temporal dependencies, delayed reactions, and volatility clustering, all of which become more pronounced during periods of uncertainty. However, despite these improvements in forecast error, its out-of-sample R^2values remain relatively modest and are often close to zero. This suggests that while the LSTM captures short-horizon nonlinear patterns more effectively than ARIMAX, the standalone neural network still offers limited explanatory power when markets are driven by overlapping macro-financial, geopolitical, and policy shocks.

At the same time, the LSTM performs particularly well in the full test period and remains highly competitive during the COVID-19 shock, where its flexibility appears especially valuable. The COVID-19 crisis was characterized by an abrupt exogenous collapse followed by unusually rapid policy intervention, liquidity support, and market rebound. Such a crisis produced highly nonlinear but relatively compressed dynamics, which favored a model capable of quickly adapting to changing temporal patterns. In this setting, the LSTM captures the sharp but short-lived nonlinear adjustments effectively, explaining why its performance is very close to the Hybrid model and why the Hybrid's incremental advantage is relatively small.

By contrast, the Hybrid model provides the most robust and generally strongest forecasting performance across the majority of crisis regimes. It achieves the lowest forecast errors in nearly all crisis periods and is the only model that consistently generates positive out-of-sample R^2, indicating not only lower forecast error but also stronger predictive content relative to the benchmark. The Hybrid model performs especially well during the Global Financial Crisis (GFC), the Eurozone sovereign debt crisis, the oil price collapse, the inflation tightening period, and the two major geopolitical conflict episodes.

The strength of the Hybrid model becomes clearer when considering the distinct nature of these crises. The GFC was a prolonged systemic financial collapse driven by leverage, banking fragility, and cascading balance-sheet effects. Such an environment involves both persistent macro-financial relationships and highly nonlinear transmission mechanisms, making it difficult for either a purely linear or purely nonlinear model to perform optimally in isolation. The Hybrid model is better suited to this setting because it combines the ARIMAX model's ability to capture structured factor relationships with the LSTM's ability to model nonlinear propagation and volatility persistence.

A similar logic applies to the Eurozone sovereign debt crisis, which was characterized by sovereign credit fragmentation, prolonged policy uncertainty, and recurrent financial stress. This was not a single abrupt shock, but rather a drawn-out regime of uncertainty and repricing. In such conditions, a Hybrid framework is particularly effective because it can jointly account for systematic macro-financial information and the evolving nonlinear market response. Likewise, during the oil price collapse, where markets were affected by global demand concerns, commodity-market instability, and cross-asset spillovers, the Hybrid model again provides the strongest performance. This suggests that crises involving multi-channel transmission mechanisms benefit from models that combine structured and adaptive forecasting components.

The Hybrid model also performs best during the inflation tightening period, which represents one of the clearest examples of a policy-driven structural shock. This episode was shaped by aggressive monetary tightening, inflation persistence, uncertainty over future policy paths, and repricing across equity and bond markets. Such an environment naturally amplifies the importance of both macroeconomic factors and nonlinear market adjustments. Here, the Hybrid model's advantage is especially intuitive: the ARIMAX component captures factor exposures, liquidity effects, and policy-related signals, while the LSTM component captures delayed nonlinear responses and interactions among financial variables. This combination yields substantial gains in forecast accuracy and explanatory power.

The same pattern emerges during the Russia–Ukraine war and the Israel–Palestine conflict, both of which introduced elevated geopolitical risk, commodity and energy uncertainty, and broad market repricing. These crises differ from financial and inflationary shocks in origin, but they similarly generate persistent uncertainty, regime shifts, and heterogeneous investor responses. The Hybrid model's continued superiority in these settings suggests that its strength lies not in fitting one specific type of crisis, but in its ability to remain robust across crises with different transmission mechanisms. In other words, although the crises in Table S5 differ substantially in their underlying nature—ranging from systemic financial collapse to sovereign debt stress, commodity disruption, inflationary tightening, and geopolitical conflict—the Hybrid model remains the most stable and generally best-performing specification because it can integrate both structured economic signals and nonlinear market behavior.

These findings align with prior forecasting research showing that hybrid models perform particularly well during periods of instability and structural breaks, when no single modeling framework is sufficient to capture the full range of market dynamics (Zhang, 2003; Timmermann, 2006). They also reinforce the interpretation that return predictability is fundamentally regime-dependent**.** Linear models perform poorly under crisis conditions, nonlinear models offer meaningful improvements—especially during fast-moving and highly nonlinear shocks—and hybrid models provide the strongest and most stable forecasting performance when crises involve persistent structural change, policy uncertainty, and multiple interacting transmission channels.

The results also support the broader interpretation that the predictive relevance of Climate Policy Uncertainty (CPU**)** is not constant over time but becomes more informative during stressed market conditions. Crises such as the inflation tightening episode and geopolitical conflicts amplify investor sensitivity to policy-related uncertainty, including uncertainty around transition risks, energy security, and regulatory responses. This is consistent with prior evidence that climate-related uncertainty exerts stronger predictive effects during periods of elevated macro-financial and policy stress (Liang et al., 2022; Raza et al., 2024). More broadly, these results suggest that the forecasting contribution of CPU depends critically on both market regime and model architecture, rather than operating as a uniformly informative predictor across all periods.

Overall, the evidence from Table S5 suggests that no single model dominates for exactly the same reason in every regime, but that the Hybrid model is the most reliable and broadly superior framework across crisis environments, while the LSTM remains highly competitive in normal periods and in short-lived, highly nonlinear shocks such as COVID-19. This distinction is important because it shows that forecasting gains are context-specific rather than universal, supporting the view that machine-learning and hybrid approaches should be understood as complements to, rather than replacements for, traditional econometric benchmarks. Given these systematic differences in predictive performance, the next step is to formally assess whether the observed gains are statistically significant using Diebold–Mariano and Clark–McCracken forecast comparison tests.

4.4 Diebold–Mariano Tests of Forecast Accuracy

Sections 4.2 and 4.3 document systematic differences in forecast performance (RMSE, MAE, out-of-sample R^2, but formal inference is needed to determine whether these differences are statistically significant. We therefore use Diebold–Mariano (DM) tests to compare the best-performing ARIMAX, LSTM, and Hybrid models under a squared-error loss. The DM framework is well suited here because it uses the time series of loss differentials to test the null of equal predictive accuracy while allowing for serial dependence in forecast errors (Diebold & Mariano, 1995). We report pairwise DM results for the full test sample (a benchmark under normal conditions) and for each crisis-period window (to assess robustness under stress). By

construction, positive DM statistics indicate that the second model in each pair has superior predictive performance.

Table S6: Diebold–Mariano tests of forecast accuracy (squared-error loss)

Period	DM (ARIMAX vs LSTM)	p-value	DM (ARIMAX vs Hybrid)	p-value	DM (LSTM vs Hybrid)	p-value
Whole period (test)	**5.69*** **	**0.000**	**8.24*** **	**0.000**	**–2.22** **	**0.026**
GFC	1.05	0.293	**4.34*** **	**0.000**	**7.64*** **	**0.000**
Eurozone Debt	0.95	0.343	**8.59*** **	**0.000**	**5.06*** **	**0.000**
Oil Collapse	**5.01*** **	**0.000**	**8.09*** **	**0.000**	**4.22*** **	**0.000**
COVID-19	**5.93*** **	**0.000**	**6.03*** **	**0.000**	1.05	0.295
Inflation Shock	**14.36*** **	**0.000**	**14.56*** **	**0.000**	**2.56***	**0.011**
Russia–Ukraine	**8.95*** **	**0.000**	**9.69*** **	**0.000**	**2.55***	**0.011**
Israel–Palestine	**13.87*** **	**0.000**	**14.02*** **	**0.000**	1.64	0.100

Notes: *The DM test evaluates* $H_0: \mathbb{E}[d_t] = 0$*, where* d_t*is the loss differential between two competing forecasts using squared-error loss. Positive DM statistics indicate that the second model in the comparison has lower expected loss.The Diebold–Mariano (DM) test evaluates the null hypothesis of equal predictive accuracy using squared-error loss. Positive statistics indicate that the second model has lower expected forecast error. *** denotes statistical significance at the 1% level (and therefore** & * also at 5% and 10%).*

The full-sample evaluation, representing periods without financial turmoil, establishes the benchmark ranking. The null of equal predictive accuracy is decisively rejected for both ARIMAX–LSTM and ARIMAX–Hybrid, showing that the nonlinear and ensemble models outperform the linear benchmark under normal conditions. The negative test statistic for the LSTM–Hybrid comparison indicates that the Hybrid model is statistically more accurate than LSTM, corroborating the full-sample results in Section 4.2. This supports the view that combining linear structure with nonlinear learning delivers additional forecasting gains even in tranquil markets. The Diebold–Mariano test results for the crisis subsamples reinforce these conclusions. The DM test results provide further evidence that observed differences in forecast accuracy are not uniform across models and periods. While statistically significant improvements are detected in selected cases, particularly during crisis regimes, the results do not support the dominance of any single forecasting approach across all horizons. This outcome is consistent with prior comparative forecasting studies emphasizing that relative model performance depends on the interaction between predictor sets, model architecture, and prevailing market conditions (Campbell and Thompson, 2008; Coulombe et al., 2022). Across all crisis episodes, the Hybrid specification strongly dominates the ARIMAX benchmark, consistent with linear forecasting relationships breaking down under regime shifts, heightened volatility, and rapid movements in risk premia. The LSTM model generally outperforms ARIMAX during crises, indicating that nonlinear and regime-sensitive dynamics are essential for accurate prediction in turbulent periods.

The LSTM–Hybrid comparison is more mixed. In several episodes—such as the Eurozone sovereign debt crisis, the oil price collapse, the inflation shock, and the Russia–Ukraine conflict—the Hybrid model substantially outperforms the LSTM, suggesting that combining systematic, factor-based structure with nonlinear adaptive components is particularly valuable when dislocations are persistent and multidimensional. By contrast, during the Global Financial Crisis (GFC) and COVID-19, the performance gap between LSTM and Hybrid is not statistically significant, implying that under extremely rapid dislocations the benefits of forecast combination are dampened by high noise and transient price distortions. In no crisis period is the LSTM statistically superior to the Hybrid model. The DM-test results confirm a stable ranking of model performance consistent with Sections 4.2–4.3. The ARIMAX model is statistically dominated, while the LSTM substantially improves predictive accuracy by

capturing nonlinear dynamics. The Hybrid model performs best in both tranquil and crisis regimes. These findings support adopting the Hybrid framework as the preferred forecasting approach and guide the final conclusions on model architecture and explanatory variable selection.

4.5 Clark–McCracken Nested Forecast Comparison Tests

Table (S7–S9). Clark–McCracken Tests of Forecast Accuracy (Robustness and Nested Comparison Framework). The following tables present Clark–McCracken tests for comparing nested forecasts. The ENC-NEW statistic examines whether the expanded model offers additional predictive content beyond the benchmark, whereas the MSE-F statistic determines whether the alternative model delivers a statistically significant improvement in forecast error. These procedures are tailored for settings with nested models, in which the competing forecasts are based on overlapping information sets.

Table S7: **Clark–McCracken Tests at** 1% significance

Period	Comparison	ENC-NEW Statistic	p-value	MSE-F Statistic
Full Test	ARIMAX vs Hybrid	27.58286	0	20375.77
Full Test	LSTM vs Hybrid	5.098823	1.71E-07	49.6494
GFC	ARIMAX vs Hybrid	10.07618	0	253.3984
GFC	LSTM vs Hybrid	7.903209	0	53.40652
Eurozone Debt Crisis	ARIMAX vs Hybrid	13.92614	0	230.3212
Eurozone Debt Crisis	LSTM vs Hybrid	5.096754	0	28.22012
Oil Price Collapse	ARIMAX vs Hybrid	13.10105	0	441.5791
Oil Price Collapse	LSTM vs Hybrid	7.091241	0	77.58051
COVID-19 Shock	ARIMAX vs Hybrid	7.39555	7.04E-14	1143.16
COVID-19 Shock	LSTM vs Hybrid	2.446270	0.007217	8.667015
Inflation Shock	ARIMAX vs Hybrid	23.42472	0	6159.178
Inflation Shock	LSTM vs Hybrid	4.650097	0.000002	26.82939
Russia–Ukraine War	ARIMAX vs Hybrid	11.34776	0	513.7809
Russia–Ukraine War	LSTM vs Hybrid	3.966788	0.000036	20.06685
Israel–Palestine Conflict	ARIMAX vs Hybrid	14.7945	0	3269.525
Israel–Palestine Conflict	LSTM vs Hybrid	4.682324	0.000001	23.58402

At the 1% significance level, the results strongly support the superiority of the Hybrid model over both ARIMAX and LSTM across all periods. The ENC-NEW statistics are consistently positive and statistically significant, indicating that the Hybrid model contains additional predictive information beyond benchmark models. The MSE-F statistics further confirm that these gains translate into meaningful reductions in forecast error.

Table S8: Clark–McCracken Tests at **5% significance**

Period	Comparison	ENC-NEW Statistic	p-value	MSE-F Statistic
Full Test	ARIMAX vs Hybrid	27.56249	0	20447.89
Full Test	LSTM vs Hybrid	3.281881	0.000516	24.94025
GFC	ARIMAX vs Hybrid	10.203212	0	249.9322
GFC	LSTM vs Hybrid	17.161264	0	389.949
Eurozone Debt Crisis	ARIMAX vs Hybrid	14.716673	0	245.305
Eurozone Debt Crisis	LSTM vs Hybrid	6.891672	0	41.6046
Oil Price Collapse	ARIMAX vs Hybrid	13.103415	0	440.6778
Oil Price Collapse	LSTM vs Hybrid	4.707539	0.000001	33.34094
COVID-19 Shock	ARIMAX vs Hybrid	7.423173	0	1143.866
COVID-19 Shock	LSTM vs Hybrid	3.682127	0.000116	10.41133

Inflation Shock	ARIMAX vs Hybrid	23.415545	0	6163.226
Inflation Shock	LSTM vs Hybrid	4.509563	0.000003	21.75771
Russia–Ukraine War	ARIMAX vs Hybrid	11.300842	0	509.7333
Russia–Ukraine War	LSTM vs Hybrid	3.905001	0.000047	18.06312
Israel–Palestine Conflict	ARIMAX vs Hybrid	14.809416	0	3263.217
Israel–Palestine Conflict	LSTM vs Hybrid	5.879774	0	36.09199

The results at the 5% significance level remain fully consistent with those reported at the 1% level, confirming the robustness of the findings across alternative inference thresholds.

Table S9: Clark–McCracken Tests at **10% significance**

Period	Comparison	ENC-NEW Statistic	p-value	MSE-F Statistic
Full Test	ARIMAX vs Hybrid	27.62235	0	20541.06
Full Test	LSTM vs Hybrid	4.745933	1.04E-06	11.19327
GFC	ARIMAX vs Hybrid	10.156880	0	247.9801
GFC	LSTM vs Hybrid	10.885166	0	127.9223
Eurozone Debt Crisis	ARIMAX vs Hybrid	13.978311	0	230.4751
Eurozone Debt Crisis	LSTM vs Hybrid	17.390612	0	493.4897
Oil Price Collapse	ARIMAX vs Hybrid	13.067806	0	436.475
Oil Price Collapse	LSTM vs Hybrid	9.448883	0	147.6728
COVID-19 Shock	ARIMAX vs Hybrid	7.421922	0	1143.612
COVID-19 Shock	LSTM vs Hybrid	2.818803	0.002410	7.668316
Inflation Shock	ARIMAX vs Hybrid	23.409798	0	6167.425
Inflation Shock	LSTM vs Hybrid	4.787767	0.000001	21.39167
Russia–Ukraine War	ARIMAX vs Hybrid	11.305755	0	510.8753
Russia–Ukraine War	LSTM vs Hybrid	4.322984	0.000008	21.81318
Israel–Palestine Conflict	ARIMAX vs Hybrid	14.806878	0	3262.858
Israel–Palestine Conflict	LSTM vs Hybrid	0.711605	0.238355	0.568739

At the 10% significance level, the results remain broadly consistent, with one marginal exception observed for the LSTM–Hybrid comparison during the Israel–Palestine conflict. In this case, the null hypothesis of equal predictive accuracy cannot be rejected, suggesting that both models perform similarly under short-lived, high-frequency shock conditions. This finding aligns with the main-text interpretation that nonlinear models alone may suffice in abrupt crisis environments.

5. Conclusion and Implications

This study investigates whether climate policy uncertainty (CPU) contains incremental predictive information for short-horizon U.S. equity returns and whether its forecasting relevance depends on both market regimes and model architecture. Using daily S&P 500 data over 2000–2024, a strictly time-ordered 60%–20%–20% training–validation–test split, and a unified comparison of ARIMAX, LSTM, and Hybrid forecasts, the evidence shows that return predictability remains weak in aggregate but becomes much more structured once crisis regimes and model form are taken into account.

Several conclusions follow from the empirical evidence. First, the linear ARIMAX benchmark performs poorly in both the full sample and crisis subsamples, confirming that conventional linear return-forecasting relationships are fragile and often unstable out of sample (Welch and Goyal, 2008; Rapach et al., 2010). By contrast, the standalone LSTM delivers the strongest performance over the full test period and remains particularly competitive in abrupt, high-frequency shock environments. This result is consistent with the broader evidence that

machine-learning architectures are better able to accommodate nonlinear interactions, temporal instability, and state dependence in financial returns (Gu et al., 2020; Bianchi et al., 2021).

Second, the predictive relevance of CPU is clearly regime-dependent rather than constant through time. In tranquil conditions, CPU adds only limited incremental forecasting value beyond standard factor, liquidity, and technical predictors. During crisis regimes, however, CPU becomes materially more informative, especially when markets are exposed to persistent macro-financial stress, policy repricing, and heightened transition-risk uncertainty. This interpretation is consistent with recent climate-finance evidence showing that climate policy uncertainty propagates through financial markets via liquidity creation, interconnectedness, and extreme-risk spillovers, with stronger effects under stressed conditions (Naifar, 2024; Xu et al., 2024; Yang et al., 2025).

Third, model architecture plays a central role in extracting CPU-related information. The combined error metrics, the Diebold–Mariano results, and the Clark–McCracken nested forecast comparisons together show that the Hybrid framework is the most robust specification across the majority of crisis periods, especially in the Global Financial Crisis, Eurozone sovereign debt crisis, oil price collapse, inflation shock, and Russia–Ukraine war. These episodes involved persistent structural disruption, overlapping transmission channels, and prolonged repricing, conditions under which combining linear macro-financial structure with nonlinear adaptive learning improves forecast stability and informational efficiency (Timmermann, 2006; Giacomini and Rossi, 2010; Clark and McCracken, 2001, 2005). At the same time, the results also show that no single model dominates for exactly the same reason in every regime: the LSTM remains strongest in the full sample and highly competitive during abrupt volatility episodes such as COVID-19, while the Israel–Palestine conflict represents a marginal case in which LSTM and Hybrid forecasts are effectively comparable under weaker significance thresholds.

These findings carry several implications. For investors and risk managers, they suggest that forecasting systems should be state-contingent: nonlinear architectures may be sufficient in tranquil or sharp transitory shock environments, whereas hybrid frameworks are more reliable when markets face persistent structural instability. For policymakers, the results imply that uncertainty surrounding climate policy can amplify market sensitivity precisely when financial conditions are already fragile, reinforcing the importance of policy clarity, consistency, and credibility in periods of macroeconomic and geopolitical stress. More broadly, the results indicate that CPU should not be treated as a uniformly informative predictor; its value depends on both the prevailing regime and the modeling framework used to extract its signal.

The scope for future research remains substantial. The present analysis focuses on a single aggregate equity index and a one-day-ahead forecasting horizon, which naturally limits generalizability. Future work could extend the framework to sectoral indices with heterogeneous carbon exposure, alternative forecast horizons, cross-sectional return prediction, or adaptive combination schemes in which forecast weights vary across market states. Incorporating additional uncertainty measures alongside CPU may also improve regime identification and help distinguish climate-policy effects from broader macro-financial uncertainty. Overall, the evidence supports a regime-aware view of return forecasting in which CPU is economically meaningful primarily during stressed market conditions and is most effectively exploited through flexible nonlinear and hybrid forecasting architectures.

The scope for *future research* remains substantial. The present analysis focuses on a single aggregate equity index at a time, which constrains the ability to generalize or to generate highly accurate predictive statements. Subsequent work could investigate alternative forecast horizons; sectoral equity indices characterized by heterogeneous levels of carbon exposure; cross-sectional return predictability; and state-dependent weighting schemes within hybrid model architectures. Moreover, incorporating a broader set of uncertainty indicators may enhance regime identification and contribute to greater stability and robustness of forecast performance.

All supplementary results support the main-text conclusions regarding regime-dependent predictability and the superior robustness of hybrid forecasting models.

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